PART I

Category theory

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Chapter 1

Categories, morphisms and duality

In this chapter we begin by defining the main objects of category theory, that is: categories, and give a few examples. We also study how we can generalize known special types of morphisms and objects into the more general setting of categories.

The main references for this chapter are [**riehlCategoryTheoryContext**; **maclaneCategoriesWorkingMath**
aluffiAlgebraChapter02009].

1.1 Categories

Definition 1.1.1 (Category). A **category** C consists of:

- a class $\mathrm{Ob}(\mathsf{C})$ of **objects**.
- for each pair of objects $X, Y \in \mathrm{Ob}(\mathsf{C})$ a collection of sets $\mathrm{Hom}_{\mathsf{C}}(X, Y)$ of **morphisms from** X **to** Y.
- for each object $X \in \mathrm{Ob}(\mathsf{C})$ a morphisms $\mathbb{1}_X \in \mathrm{Hom}_{\mathsf{C}}(X, X)$ called the **identity** morphism.
- for each pair of morphisms $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ and $g \in \mathrm{Hom}_{\mathsf{C}}(Y, Z)$ a morphism $g \circ f \in \mathrm{Hom}_{\mathsf{C}}(X, Z)$ called their **composite**.

satisfying the following axioms:

- Composition is associative: for each quadruple $X, Y, Z, W \in \mathrm{Ob}(\mathsf{C})$ of objects, if $f \in \mathrm{Hom}_{\mathsf{C}}(Y, Z)$, $g \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ and $h \in \mathrm{Hom}_{\mathsf{C}}(W, X)$ then

$$(f \circ g) \circ h = f \circ (g \circ h)$$

- Composition satiesfies the unit laws: for each pair of objects $X, Y \in \mathrm{Ob}(\mathsf{C})$ if $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ then

$$\mathbb{1}_Y \circ f = f = f \circ \mathbb{1}_X$$

Classical set theoretical problems, such as Russell's paradox that asserts that there cannot be set of all sets, force us to define the collection of objects in a category as a proper class and not a set.

In some cases the collection of objects is small enough to be a set. We call this types of categories small.

Definition 1.1.2 (Small category). A category C is **small** if its collection of objects $\mathrm{Ob}(\mathsf{C})$ is a set.

Remark 1.1.3 (Size issues). The above definition of a category is actually one of a *locally small* category.
In the general setting, the collection of morphisms between two objects need not to form a set because it is *too big*. There are several ways to fix this: working with proper classes, with Grothendieck universes, ...
We will however be sketchy about this kind of foundational issues. Instead, will refer as "big" categories those whose collection of morphism does not form a set and not bother with the precise definition of what this actually means. By doing this we refrain ourselves to study this kind of categories, but as it turns at this will not be needed.
For the interested reader, **[riehlCategoryTheoryContext]** provides an account of why this issues may be important in certain situations.

Notation. To indicate that X is an object of a category C we write

$$X \in \mathsf{C} \text{ instead of } X \in \mathrm{Ob}(\mathsf{C})$$

To indicate that $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ is a morphism from X to Y we write

$$f : X \to Y \text{ or } X \xrightarrow{f} Y$$

We will sometimes abbreviate the composition $f \circ g$ as fg.

Before going further, we give some examples of categories that the reader may be used to.

Example 1.1.4.

- The category Set has the collection of all sets as objects and mappings as morphisms.
- The category Grp has groups as objects and group homomorphisms as morphisms.
- The category Ab has abelian groups as objects and group homomorphisms as morphisms. It is a full subcategory of Grp.
- The category Ring has rings as objects and ring homomorphisms as morphisms.[a]
- The category Rng has (not necessarilly unital) rings as objects and ring homomorphisms as morphism.
- The category Top has topological spaces as objects and continuous functions as morphisms.
- For a fixed ring R, the category $\mathsf{Mod}(R)$ has (left) R-modules as objects and R-linear maps as morphisms.
- For a fixed field k, the category Vect_k has k-vector spaces as objects and k-linear maps as morphisms.

[a] We adopt the convetion that all rings have an unit and all ring homomorphisms preserve it.

Example 1.1.5 (Category of open sets). Let X be a topological space. There is a category of open sets $\mathsf{Open}(X)$ consisting of

- As objects, the open subsets of X.
- As morphisms, inclusion of open sets: $U \to V$ if and only if $U \subseteq V$

Example 1.1.6 (Discrete category). Given a category C we can define the **discrete category** C_d taking the same collection of objects as C but only the identity morphisms between them.

Often times when studying a certain type of objects it is useful (or even necessary) to restrict our attention to an smaller but more well-behaved subclass. For example, when studying groups one may want to only consider abelian groups or when studying rings to stay in the unital case.

This notion is formalized using subcategories.

Definition 1.1.7 (Subcategory). Given a category C, a **subcategory** D consists of a subcollection of objects and a subcollection of morphisms in C such that

- If $f : X \to Y$ is a morphism in D then X, Y are objects in D.
- If $f : X \to Y$ and $g : Y \to Z$ are morphisms in D, the composition $gf : X \to Z$ is in D.
- If X is an object in D then the identity map $\mathbb{1}_X$ is in D.

We say that D is a **full subcategory** if for all objects X, Y in D every morphism $f : X \to Y$ in C is also in D.

Example 1.1.8.

- Ab is a full subcategory of Grp.
- The category of finite sets FinSet is a full subcategory of Set.
- Ring is a subcategory of Rng that is not full.
- If k is a field, Vect_k is a full subcategory of $\mathsf{Mod}(k)$.

A similar situation happens often when one wants to combine multiple objects into a single, bigger one. A simple way to do this is to consider their "product", for example by considering the cartesian product of two sets or the product topology of two topological spaces. Even though we will introduce later what the categorical product of two objects in a category is, we define it now for categories.

Definition 1.1.9 (Product category). Let C, D be two categories. The **product category** $\mathsf{C} \times \mathsf{D}$ is the category consisting of

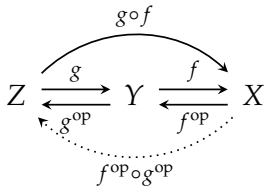
- As objects: ordered pairs (C, D), where C is an object in C and D is an object in D, i.e:
$$\mathrm{Ob}(\mathsf{C} \times \mathsf{D}) := \mathrm{Ob}(\mathsf{C}) \times \mathrm{Ob}(\mathsf{D})$$
- As morphisms: ordered pairs $(C \xrightarrow{f} C', D \xrightarrow{g} D')$, where $C \xrightarrow{f} C'$ is a morphism in C and $D \xrightarrow{g} D'$ is a morphism in D, i.e: for all objects $(C, D), (C', D')$ in $\mathsf{C} \times \mathsf{D}$
$$\mathrm{Hom}_{\mathsf{C}\times\mathsf{D}}((C, D), (C', D')) := \mathrm{Hom}_{\mathsf{C}}(C, C') \times \mathrm{Hom}_{\mathsf{D}}(D, D')$$
- Composition is defined *componentwise*: if $(C \xrightarrow{f} C', D \xrightarrow{g} D')$ and $(C' \xrightarrow{f'} C'', D' \xrightarrow{g'} D'')$ are two morphisms in $\mathsf{C} \times \mathsf{D}$ then
$$(f', g') \circ (f, g) := (f' \circ f, g' \circ g) : C \times D \to C'' \times D''$$

1.1.1 Duality

In the definition of a morphism in a category one has to make a choice of a source and a target. A natural question is to consider what happens if we reverse this arrows. As it turns out, this produces another category that is dual to the original one.

Definition 1.1.10 (Opposite category). Let C be a category. The **opposite category** C^{op} consists of:

- The same objects as C: $\mathrm{Ob}(\mathsf{C}^{\mathrm{op}}) := \mathrm{Ob}(\mathsf{C})$.
- For each pair of objects $X, Y \in \mathsf{C}^{\mathrm{op}}$: $\mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(X, Y) := \mathrm{Hom}_{\mathsf{C}}(Y, X)$.
- If $f^{\mathrm{op}} \colon X \to Y$, $g^{\mathrm{op}} \colon Y \to Z$ are morphisms in C^{op} their composition is defined by
$$f^{\mathrm{op}} \circ g^{\mathrm{op}} := (g \circ f)^{\mathrm{op}}$$
where $f \colon Y \to X$ and $g \colon Z \to Y$ are the associated morphisms in C. Visually



Example 1.1.11. Consider a preordered set $(P, \leq)$ viewed as a small category where a unique morphism $x \to y$ exists if and only if $x \leq y$.
The opposite category P^{op} corresponds to the same set P equipped with the reversed ordering $\geq$. That is, a morphism $x \to y$ exists in P^{op} if and only if $y \leq x$ in P.

Remark 1.1.12. Let $f^{\mathrm{op}} \colon Y \to X$ and $g^{\mathrm{op}} \colon Y \to X$ be two morphisms in the opposite category C^{op}. Then $f^{\mathrm{op}} = g^{\mathrm{op}}$ in C^{op} if and only if $f = g$ in C.

A categorical statement P^* is defined as the **dual** of a statement P in a category C if P^* holds in the opposite category C^{op} if and only if P holds in C.

Note that, by definition, for any category C we have that $(\mathsf{C}^{\mathrm{op}})^{\mathrm{op}} = \mathsf{C}$.

Therefore if a statement is true for all categories C, its dual is true for all categories C^{op}. Since C^{op} is an arbitrary category the dual statement is also true for all categories.

In practice, this means that proving an statement directly proves its dual for free.

1.2 Types of morphisms

When starting to learn about set theory, some of the first definitions are those of an injection and a surjection. Then when studying algebraic objects such as groups and rings one learns about monomorphisms and epimorphisms, that are no more and no less than injections and surjections between the underlying sets of the objects that respect the desired algebraic operation.

To translate it to the general categorical framework we cannot speak of objects and thus resort to an equivalent definition.

Definition 1.2.1 (Monomorphism and epimorphism). Let $f : X \to Y$ be a morphism in a category C. We say that f is a

- **Monomorphism** (or is **monic**) if for every pair of parallel morphisms $g, h : Z \to X$ in C, the equality $f \circ g = f \circ h$ implies $g = h$, i.e: f is left-cancellative.
- **Epimorphism** (or is **epic**) if for every pair of parallel morphisms $g, h : Y \to Z$ in C, the equality $g \circ f = h \circ f$ implies $g = h$, i.e: f is right-cancellative.

Example 1.2.2.

- In Set, an monomorphism (resp. epimorphism) is an injective (resp. surjective) function.
- In Grp, an monomorphism (resp. epimorphism) is an injective (resp. surjective) group homomorphism.
- In Top, an monomorphism (resp. epimorphism) is an injective (resp. surjective) continuous map.

One *caveat* of this definition is that the usual definition of a ring epimorphism (i.e: a surjective ring homomorphism) is too restrictive.

Example 1.2.3. In Ring, there are epimorphisms that are not surjective ring homomorphisms.
For example, consider the inclussion $\iota : \mathbb{Z} \hookrightarrow \mathbb{Q}$. For any ring homomorphism $f : \mathbb{Q} \to R$, where R is any ring, since for all $\frac{a}{b} \in \mathbb{Q}$ we have $a, b \in \mathbb{Z}$

$$f\left(\frac{a}{b}\right) = f(a)f(b^{-1}) = f(a)f(b)^{-1}$$

Therefore f is completely determined by its values in $\mathbb{Z}$ and, since $f \circ \iota = f|_{\mathbb{Z}}$ we get that ι is right-cancellative.

As an example of the duality phenomena we described earlier, we show that monomorphisms and epimorphisms are dual concepts.

Lemma 1.2.4 (Monomorphisms and epimorphisms are dual). Let $f : X \to Y$ be a morphism in a category C. Consider the associated morphism $f^{\mathrm{op}} : Y \to X$ in the opposite category C^{op}.
Then f is a monomorphism (resp. epimorphism) in C if and only if f^{op} is an epimorphism (resp. epimorphism) in C^{op}.

Proof. Consider of pair of morphisms $g, h : Z \rightrightarrows X$ in C and their opposites $g^{\mathrm{op}}, h^{\mathrm{op}} : X \rightrightarrows Z$ in C^{op}.

Note that, by definition of composition in C^{op} we have

$$g^{\mathrm{op}} \circ f^{\mathrm{op}} = (f \circ g)^{\mathrm{op}}$$

Therefore we have that $f \circ g = f \circ h$ implies $g = h$ if and only if $(f \circ g)^{\mathrm{op}} = (f \circ h)^{\mathrm{op}}$ implies $g = h$ if and only if $g^{\mathrm{op}} \circ f^{\mathrm{op}} = h^{\mathrm{op}} \circ f^{\mathrm{op}}$ implies $g^{\mathrm{op}} = h^{\mathrm{op}}$. □

Remark 1.2.5. In what follows we will not spell out the proofs of two objects being dual. The proof strategy is basically the same as in the above lemma in all cases.

Another recurrent notion accross mathematics is that of an isomorphism. In short, we say that two objects are isomorphic if they *behave* functionally the same even if

they *look* different.

We can extend this notion to general categories.

Definition 1.2.6 (Isomorphism). A morphism $f : X \to Y$ in a category C is an **isomorphism** if there exists another morphism $g : Y \to X$ in C such that

$$g \circ f = \mathbb{1}_X \quad \text{and} \quad f \circ g = \mathbb{1}_Y$$

The morphism g is called the **inverse** of f and is denoted by f^{-1}.

Example 1.2.7.

- In Set, an isomorphism is a bijection.
- In Grp, an isomorphism is a bijective group homomorphism.
- In Top, an isomorphism is a homeomorphism.
- In Ring, an isomorphism is a bijective ring homomorphism.

The usual properties of the inverses and isomorphisms work as one would expect in the general setting.

Proposition 1.2.8. Let $f : X \to Y$ and $g : Y \to Z$ be isomorphisms in a category C. Then:

1. The inverse f^{-1} of f is unique.
2. The identity $\mathbb{1}_X$ is an isomorphism for every $X \in \mathsf{C}$.
3. $g \circ f$ is an isomorphism with $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.
4. If $g \circ f$ is an isomorphism and f is an isomorphism, then g is an isomorphism. Analogously if $g \circ f$ and g are isomorphisms, then f is an isomorphism.

Proof.

1. Suppose that $h_1, h_2 : Y \to X$ are inverses of f. We have
$$h_1 = h_1 \circ \mathbb{1}_Y = h_1 \circ (f \circ h_2) = (h_1 \circ f) \circ h_2 = \mathbb{1}_X \circ h_2 = h_2$$
2. For all $X \in \mathsf{C}$ by definition
$$\mathbb{1}_X \circ \mathbb{1}_X = \mathbb{1}_X$$
i.e: $\mathbb{1}_X$ is its own left and right inverse.
3. We only need to check that $f^{-1} \circ g^{-1}$ is both a left and right inverse of $g \circ f$. On the one hand
$$(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ f = \mathbb{1}_X$$
On the other
$$(g \circ f) \circ (f^{-1} \circ g^{-1}) = g \circ (f \circ f^{-1}) \circ g^{-1} = g \circ g^{-1} = \mathbb{1}_Z$$
4. Let $h := (g \circ f)^{-1} : Z \to X$. We are going to check that $f \circ h : Z \to Y$ is the inverse of g. On the one hand
$$g \circ (f \circ h) = (g \circ f) \circ h = \mathbb{1}_Z$$
On the other, since $h \circ (g \circ f) = \mathbb{1}_X$ precomposing with f and postcomposing with f^{-1} we have
$$\begin{aligned} f \circ (h \circ g \circ f) \circ f^{-1} = f \circ \mathbb{1}_X \circ f^{-1} = \mathbb{1}_Y &\implies (f \circ h) \circ g \circ (f \circ f^{-1}) = \mathbb{1}_Y \\ &\implies (f \circ h) \circ g = \mathbb{1}_Y \end{aligned}$$

□

As it happens in Set, the notions of monomorphisms, epimorphisms and isomorphisms are closely related.

Proposition 1.2.9. Let $f: X \to Y$ be an isomorphism. Then f is both a monomorphism and an epimorphism.

Proof. Suppose $f: X \to Y$ is an isomorphism in a category C. Then there exists a morphism $f^{-1}: Y \to X$ in C such that

$$f^{-1} \circ f = \mathbb{1}_X \quad f \circ f^{-1} = \mathbb{1}_Y$$

Consider a pair of morphisms $g, h: Z \rightrightarrows X$ such that $f \circ g = f \circ h$. Precomposing with f^{-1} we get

$$f^{-1} \circ (f \circ g) = f^{-1} \circ (f \circ h) \implies (f^{-1} \circ f) \circ g = (f^{-1} \circ f) \circ h \implies g = h$$

Therefore f is a monomorphism.

Analogously, if we consider pair of morphism $g', h': Y \rightrightarrows Z$ such that $g' \circ f = h' \circ f$ this time by postcomposing with f^{-1} we get $g' = h'$, i.e: f is a epimorphism. □

Example 1.2.10. In general, a morphism $f: X \to Y$ that is both a monomorphism and epimorphism is not an isomorphism.
Indeed, in the category Top of topological spaces consider a set X with more than one point endowed with the discrete X_d and trivial X_t topology.
Consider the identity map $\mathbb{1}_X: X_d \to X_t$, which is continuous. Since $\mathbb{1}_X$ is injective we have that it is a monomorphism and, since it is surjective, it is an epimorphism.
However since $id_X^{-1}: X_t \to X_d$ is not continuous, $\mathbb{1}_X$ cannot be an isomorphism.

As we have seen before, one may create an smaller subcategory by restricting their attention to a smaller collection of objects. In this cases, isomorphisms in the smaller category are preserved the bigger one.

Proposition 1.2.11. Let D be a subcategory of a category C. If $f: X \to Y$ is an isomorphism in D then f is also an isomorphism in C.

Proof. Since $f: X \to Y$ is an isomorphism in D there exists an inverse $f^{-1}: Y \to X$ in D. Therefore since $\mathrm{Hom}_{\mathsf{D}}(Y, X) \subseteq \mathrm{Hom}_{\mathsf{C}}(Y, X)$, f^{-1} is an inverse of f in C. □

If the subcategory is full we also have the reciprocal.

Proposition 1.2.12. Let D be a full subcategory of a category C. If $f: X \to Y$ is an isomorphism in C and $X, Y \in \mathsf{D}$ then f is also an isomorphism in D.

Proof. Since $f: X \to Y$ is an isomorphism in C there exists an inverse $f^{-1}: Y \to X$ in D. Since D is a full subcategory of D and $X, Y \in \mathsf{D}$ we have that $f^{-1} \in \mathrm{Hom}_{\mathsf{D}}(Y, X)$. Therefore f admits an inverse in D. □

1.3 Special objects

Just like in a category there are special types of morphisms, we also have special objects. For example, every group has a homomorphism from the trivial group and

every unital ring has a homomorphism into the ring of integers. This are examples of what are called initial and terminal objects.

Definition 1.3.1 (Initial and terminal objects). An object $I \in \mathsf{C}$ is **initial** if for every object $X \in \mathsf{C}$ there is a unique morphism $I \to X$.
Dually, an object $T \in \mathsf{C}$ is **terminal** if for every object $X \in \mathsf{C}$ there is a unique morphism $X \to T$.

Example 1.3.2.

- In Set, the empty set $\varnothing$ is initial and the singleton $\{*\}$ is terminal.
- In Grp, trivial groups are both initial and terminal.
- In both Ring and Rng, the zero ring is initial. However, the terminal is different: in Ring it is $\mathbb{Z}$ and in Rng is the zero ring.
- In Top, the empty topological space is initial and the singleton $\{*\}$ is terminal.
- There are no initial nor terminal objects in Field, since there are no homomorphisms between fields of different characteristic.

In the above examples we have spoken of *the* initial and *the* terminal instead of *an* initial or *a* terminal object. This is justified as this types of objects are isomorphic.

Proposition 1.3.3 (Uniqueness of initial and terminal objects). Let X, Y be two initial (resp. terminal) objects in a category C. Then there exists a unique isomorphism $X \xrightarrow{\sim} Y$ in C.

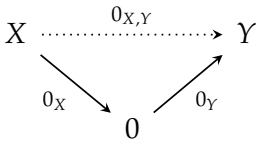
Proof. Since X is initial in C there exists a unique morphism $f : X \to Y$. Analogously there exist a unique morphism $g : Y \to X$.

Note that $g \circ f : X \to X$. Therefore since X is initial $g \circ f = \mathbb{1}_X$. Analogously since $f \circ g : Y \to Y$ we have $f \circ g = \mathbb{1}_Y$.

Therefore f and g are inverses of each other, i.e: $X \cong Y$. Moreover they are unique by definition. □

In some particular cases initial and terminal objects coincide, which means that there is an object with a unique morphism from and to every other object in the category. In this case we say that this object is zero.

Definition 1.3.4 (Zero object). An object $0 \in \mathsf{C}$ is the **zero** object if it is both initial and terminal.
Note that for any two objects $X, Y \in \mathsf{C}$ there exist unique maps $0_X : X \to 0$ and $0_Y : 0 \to Y$. We call this composite map the **zero map** between X and Y and denote it $X \xrightarrow{0_{X,Y}} Y$:



By abuse of notation, we will denote all of these maps by 0.

Example 1.3.5.

- Set and Top have no zero objects since $\varnothing \neq \{*\}$.
- In Grp and Ab, the zero group is the zero object.
- Ring has no zero objects. However in Rng the zero ring is the zero object.

As before, we can speak of *the* zero object (up to isomorphism).

Corollary 1.3.6 (Uniqueness of zero objects)**.** Let $0, \widetilde{0}$ be two zero objects in a category C. Then there exists a unique isomorphism $0 \xrightarrow{\sim} \widetilde{0}$ in C.

Proof. Since 0 and $\widetilde{0}$ are in particular initial we can apply Proposition 1.3.3. □

Chapter 2

Functors and natural transformations

In this chapter we study functors, which encode how we move data across different categories. We also introduce the concept of natural transformations, which in turn encode now moving data across functors, and equivalence of categories.

The main references for this chapter are [**riehlCategoryTheoryContext**; **maclaneCategoriesWorkingMath**
aluffiAlgebraChapter02009].

2.1 Functors

As we have seen, category theory tries to study objects by means of their transformations. Since we can consider categories as their own mathematical object a natural question arises: how do we model transformations between categories?

The answer to this question are functors.

Definition 2.1.1 (Functor). Let C, D be two categories. A (covariant) **functor** $F : \mathsf{C} \to \mathsf{D}$ consists of:

- for each object $X \in \mathsf{C}$, an object $F(X) \in \mathsf{D}$.
- for each morphism $f : X \to Y$ in C, a morphism $Ff : F(X) \to F(Y)$ in D,

such that:

- $F(\mathbb{1}_X) = \mathbb{1}_{F(X)}$ for every $X \in \mathsf{C}$.
- $F(g \circ f) = Fg \circ Ff$ for every composable pair of morphisms f, g in C.

A functor is **contravariant** if it reverses the order of composition: $F(g \circ f) = Ff \circ Fg$.
This is equivalent to a covariant functor $F : \mathsf{C}^{\mathrm{op}} \to \mathsf{D}$.
Diagramatically, a covariant functor is represent by

$$\begin{array}{ccc} F\colon \mathsf{C} & \longrightarrow & \mathsf{D} \\ X & & F(X) \\ {\scriptstyle f}\downarrow & \longmapsto & \downarrow{\scriptstyle Ff} \\ Y & & F(Y) \end{array}$$

And a contravariant functor by

$$\begin{array}{ccc} F\colon \mathsf{C} & \longrightarrow & \mathsf{D} \\ X & & F(X) \\ f\downarrow & \longmapsto & \uparrow Ff \\ Y & & F(Y) \end{array}$$

Remark 2.1.2. We will disfavour the covariant/contravariant distinction. Therefore all functors will be covariant (unless otherwise specified) and contravariant functors will be covariant functors from the opposite category.

Example 2.1.3.

- There is a (covariant) endofunctor $P\colon \mathsf{Set} \to \mathsf{Set}$ that sends a set A to its power set $PA = \{A' \subseteq A\}$ and a map $f\colon A \to B$ to the direct image $f_*\colon PA \to PB$, $f_*(A') := f(A')$.
  We also get a contravariant version $P\colon \mathsf{Set}^{\mathrm{op}} \to \mathsf{Set}$ that sends a map $f\colon A \to B$ to the inverse image $f^{-1}\colon PB \to PA$, $f^{-1}(B') := f^{-1}(B)$.
- There is a functor $F\colon \mathsf{Set} \to \mathsf{Grp}$ that sends a set X to the free group on X.
- There is a functor $-^*\colon \mathsf{Vect}_{\mathbb{K}}^{\mathrm{op}} \to \mathsf{Vect}_{\mathbb{K}}$ that sends a vector space V to its dual V^*.
- There is a functor $\mathrm{Spec}\colon \mathsf{CRing}^{\mathrm{op}} \to \mathsf{Top}$ that sends a commutative ring R to its set $\mathrm{Spec}(R)$ of prime ideals endowed with the Zariski topology.

Example 2.1.4 (Identity functor). Let C be a category. There exists an identity functor

$$\begin{array}{ccc} \mathbb{1}\colon \mathsf{C} & \longrightarrow & \mathsf{C} \\ X & & X \\ f\downarrow & \longmapsto & \downarrow f \\ Y & & Y \end{array}$$

Example 2.1.5 (Constant functor). Let C be a category. For every object $Z \in \mathsf{C}$ there exists a constant functor

$$\begin{array}{rccc} \mathrm{ct}_Z\colon & \mathsf{C} & \longrightarrow & \mathsf{C} \\ & X & & Z \\ & f\downarrow & \longmapsto & \downarrow \mathbb{1}_Z \\ & Y & & Z \end{array}$$

Example 2.1.6 (Bifunctor). Let C, C' and D be three categories. Recall that we can construct the product category $\mathsf{C} \times \mathsf{C}'$.
A **bifunctor** is a functor of the form $F\colon \mathsf{C} \times \mathsf{C}' \to \mathsf{D}$. This means that for any $X \in \mathsf{C}$ and $X' \in \mathsf{C}$, $F(X, -)\colon \mathsf{C}' \to \mathsf{D}$ and $F(-, X')\colon \mathsf{C} \to \mathsf{D}$ are functors.
Furthermore for any morphisms $f\colon X \to Y$ in C and $g\colon X' \to Y'$ in C' we have a commutative diagram

$$\begin{array}{ccc} F(X,X') & \xrightarrow{F(\mathbb{1}_X, g)} & F(X,Y') \\ {\scriptstyle F(f,\mathbb{1}_{X'})}\downarrow & \searrow^{F(f,g)} & \downarrow{\scriptstyle F(f,\mathbb{1}_{Y'})} \\ F(Y,X') & \xrightarrow[F(\mathbb{1}_Y, g)]{} & F(Y,Y') \end{array}$$

Example 2.1.7. Let C be a category. For every object $C \in \mathsf{C}$ there exists a *covariant* Hom functor

$$\begin{array}{rccc} \mathrm{Hom}(C,-)\colon & \mathsf{C} & \longrightarrow & \mathsf{Set} \\ & X & & \mathrm{Hom}(C,X) \\ & f\downarrow & \longmapsto & \downarrow f\circ \\ & Y & & \mathrm{Hom}(C,Y) \end{array}$$

and a *contravariant* Hom functor

$$\begin{array}{rccc} \mathrm{Hom}(-,C)\colon & \mathsf{C} & \longrightarrow & \mathsf{Set} \\ & X & & \mathrm{Hom}(C,X) \\ & f\downarrow & \longmapsto & \uparrow \circ f \\ & Y & & \mathrm{Hom}(C,Y) \end{array}$$

We can combine this two into a bifunctor

$$\begin{array}{rccc} \mathrm{Hom}(-,-)\colon & \mathsf{C}^{\mathrm{op}}\times\mathsf{C} & \longrightarrow & \mathsf{Set} \\ & (X,Y) & & \mathrm{Hom}(X,Y) \\ & (f,g)\downarrow & \longmapsto & \downarrow g\circ -\circ f \\ & (Z,W) & & \mathrm{Hom}(Z,W) \end{array}$$

Example 2.1.8. Let D be a subcategory of a category C. There exist an inclussion functor

$$\begin{array}{rccc} \iota\colon & \mathsf{D} & \hookrightarrow & \mathsf{C} \\ & X & & X \\ & f\downarrow & \longmapsto & \downarrow f \\ & Y & & Y \end{array}$$

Example 2.1.9 (Preimage functor)**.** Let X, Y be two topological spaces. Consider their categories of open sets $\mathsf{Open}(X), \mathsf{Open}(Y)$.
For every continuous map $f : X \to Y$ we have a preimage functor

$$\begin{array}{rccc} f^{-1}\colon & \mathsf{Open}(Y) & \longrightarrow & \mathsf{Open}(X) \\ & U & & f^{-1}(U) \\ & \downarrow & \longmapsto & \downarrow \\ & V & & f^{-1}(V) \end{array}$$

Proposition 2.1.10 (Functors preserve isomorphisms)**.** Let C, D be two categories and $F\colon \mathsf{C} \to \mathsf{D}$ a functor between them. If f is an isomorphism in C then Ff is an isomorphism in D.

Proof. Let $f : X \to Y$ be a isomorphism in C. Then there exists an inverse $f^{-1}\colon Y \to X$ in C such that

$$f^{-1} \circ f = \mathbb{1}_X \quad f \circ f^{-1} = \mathbb{1}_Y$$

Applying F to this equalities we have

$$\begin{aligned} F(f^{-1} \circ f) = F\mathbb{1}_X &\implies Ff^{-1} \circ Ff = \mathbb{1}_{F(X)} \\ F(f \circ f^{-1}) = F\mathbb{1}_Y &\implies Ff \circ Ff^{-1} = \mathbb{1}_{F(Y)} \end{aligned}$$

i.e: Ff has an inverse Ff^{-1} in D. □

Definition 2.1.11 (Functor types). A functor $F\colon \mathsf{C} \to \mathsf{D}$ is:

- **faithful** if for every pair of objects $X, Y \in \mathsf{C}$, the map $\mathrm{Hom}_{\mathsf{C}}(X,Y) \to \mathrm{Hom}_{\mathsf{D}}(F(X),F(Y))$ is injective.
- **full** if for every pair $X, Y \in \mathsf{C}$, the map $\mathrm{Hom}_{\mathsf{C}}(X,Y) \to \mathrm{Hom}_{\mathsf{D}}(F(X),F(Y))$ is surjective.
- **fully faithful** if it is both full and faithful.
- **essentially surjective** if for every object $Y \in \mathsf{D}$ there exists an object $X \in \mathsf{C}$ such that $F(X) \cong Y$.

Example 2.1.12.

- The functor $U\colon \mathsf{Grp} \to \mathsf{Set}$ that sends a group to its underlying set is faithful.
- A subcategory D of a category C is full if and only if the inclusion functor $\iota : \mathsf{D} \to \mathsf{C}$ is full.
- Given a field $\mathbb{K}$, let $\mathsf{C}_{\mathbb{K}}$ be the category whose objects are $\mathbb{K}^n$ for some $n \in \mathbb{N}$. The inclusion $\mathsf{C}_{\mathbb{K}} \to \mathsf{Vect}_{\mathbb{K}}$ is essentially surjective, since all finite dimensional $\mathbb{K}$-vector spaces are isomorphic to $\mathbb{K}^n$ for some n.

A concept that can be abstracted using these definitions is that of a category consisting of "sets with extra structure".

Definition 2.1.13 (Concrete category). A category C is **concrete** if there exists a faithful functor $U\colon \mathsf{C} \to \mathsf{Set}$. Such functor is often times called a **forgetful functor**.

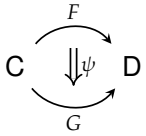
2.2 Natural transformations

We can go further and try to model now how to move data between two functors.

Definition 2.2.1 (Natural transformation). Let $F, G\colon \mathsf{C} \to \mathsf{D}$ be two functors. A **natural transformation** $\psi\colon F \implies G$ consists of a family of morphisms $\psi_X : F(X) \to G(X)$ in D, one for each object $X \in \mathsf{C}$, called the **components** of ψ, such that for every morphism $f : X \to Y$ in C, the following diagram commutes:

$$\begin{array}{ccc} F(X) & \xrightarrow{\psi_X} & G(X) \\ {\scriptstyle Ff}\downarrow & & \downarrow{\scriptstyle Gf} \\ F(Y) & \xrightarrow[\psi_Y]{} & G(Y) \end{array}$$

Diagramatically, a natural transformations is represented by



If each component ψ_X is an isomorphism in D, we say that ψ is a **natural isomorphism**, and we write $F \cong G$.

Example 2.2.2. In $\mathsf{Vect}_{\mathbb{K}}$, the map $\mathrm{ev}\colon V \to V^{**}$ that sends a vector $v \in V$ to the linear evaluation function $\mathrm{ev}_v\colon V^* \to \mathbb{K}$ defines the components of a natural transformation from the identity functor $\mathbb{1}\colon \mathsf{Vect}_{\mathbb{K}} \to \mathsf{Vect}_{\mathbb{K}}$ to the double dual functor $-^{**}\colon \mathsf{Vect}_{\mathbb{K}} \to \mathsf{Vect}_{\mathbb{K}}$.
If we restrict ourselves to the finite-dimensional case then this is a natural isomor-

phism.

By combining functors and natural transformation we obtain a new example of a category.

Definition 2.2.3 (Functor category)**.** Let C and D be categories. The **functor category** $\mathsf{Fct}(\mathsf{C}, \mathsf{D})$ has as objecs functors $F\colon \mathsf{C} \to \mathsf{D}$ and as morphisms natural transformations $\eta\colon F \implies G$. Composition and identities are defined componentwise.

Using the notion of natural isomorphisms one can define the notion of an **isomorphism** of categories. Two categories C and D are isomorphic (denoted $\mathsf{C} \cong \mathsf{D}$) if there exists a natural isomorphism between them.

This notion, however, is too restrictive to be useful. The correct notion to identify categories is that of an equivalence.

Definition 2.2.4 (Equivalent categories)**.** Two categories C, D are **equivalent** if there exist functors $F\colon \mathsf{C} \to \mathsf{D}$ and $G\colon \mathsf{D} \to \mathsf{C}$ such that $G \circ F \cong \mathbb{1}_{\mathsf{C}}$ and $F \circ G \cong \mathbb{1}_{\mathsf{D}}$. In this case, F (and G) is called an **equivalence of categories**. We denote this by $\mathsf{C} \simeq \mathsf{D}$.

In general proving that two equivalence of categories by directly constructing the functors F and G is not easy. Instead one can use the following theorem, which we state without proof.

Theorem 2.2.5 ([riehlCategoryTheoryContext])**.** A functor $F\colon \mathsf{C} \to \mathsf{D}$ defining an equivalence of categories is fully faithful and essentially surjective. Assuming the axiom of choice, any functor with these properties defines an equivalence of categories.

Chapter 3

Limits and colimits

In this chapter we study how different constructions across mathematics can be abstracted into particular cases of what are called limits and colimits. Once these concepts are defined, we spend the rest of the chapter giving familiar constructions that can be made from them. We end the chapter by proving an existence criterion for limits and colimits both in general and for the concrete case of functor categories.

The main references for this chapter are [**riehlCategoryTheoryContext**; **maclaneCategoriesWorkingMath**
aluffiAlgebraChapter02009].

3.1 Diagrams, cones and cocones

A common tool used in mathematics to relate objects and morphisms are diagrams such as commutative squares.

This notions can be easily abstracted using functors.

Definition 3.1.1 (Diagram)**.** Let J be a small category.
A **diagram** X in a category C with shape J is a functor $X\colon \mathsf{J} \to \mathsf{C}$. For every $j \in \mathsf{J}$ we write $X_j \coloneqq X(j)$.

Example 3.1.2 (Constant diagram)**.** A diagram $X\colon \mathsf{J} \to \mathsf{C}$ is **constant** if it is a constant functor const_c.
In this case the data of the diagram corresponds to the data given by a single object $c \in \mathsf{C}$.

Example 3.1.3. Consider the category

$$\mathsf{J} = \begin{array}{ccc} \bullet & \longrightarrow & \bullet \\ \downarrow & \searrow & \downarrow \\ \bullet & \longrightarrow & \bullet \end{array} = \begin{array}{ccc} 1 & \longrightarrow & 2 \\ \downarrow & \searrow & \downarrow \\ 3 & \longrightarrow & 4 \end{array}$$

A diagram with shape J is a commutative square

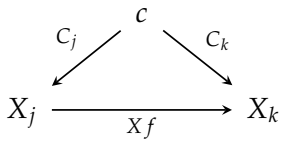
$$\begin{array}{ccc} X_1 & \longrightarrow & X_2 \\ \downarrow & & \downarrow \\ X_3 & \longrightarrow & X_4 \end{array}$$

3.1.1 Cones and cocones

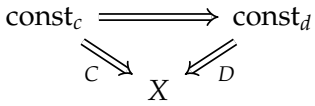
To study diagrams from the category theory point of view we must understand their transormations. In this case since diagrams are special types of functors these correspond to natural transformations.

Out of all of the possible natural transformations we will only be interested in two particular and dual cases: cones and cocones.

Definition 3.1.4 (Cone). A **cone** C over a diagram $X\colon \mathsf{J} \to \mathsf{C}$ with *apex* c is a natural transformation $C\colon \text{const}_c \Rightarrow X$ from the constant diagram to X. The components $\{C_j\colon c \to X_j\}_{j\in\mathsf{J}}$ are called the *legs* of the cone.
Less abstractly, a cone C over $X\colon \mathsf{J} \to \mathsf{C}$ with *apex* c is a collection of morphisms $C_j\colon c \to X_j$ for every object $j \in \mathsf{J}$ such that for every morphism $f : j \to k$ in J the following diagram in C commutes

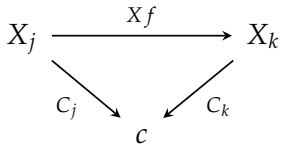


If $C\colon \text{const}_c \Rightarrow X$ and $D\colon \text{const}_d \Rightarrow X$ are two cones over X a morphism of cones is a natural transformation $\text{const}_c \Rightarrow \text{const}_d$ making the following diagram commute

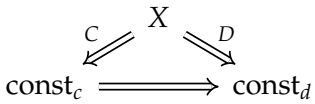


Therefore for any diagram $X : \mathsf{J} \to \mathsf{C}$ we have a category of cones $\Delta \downarrow X$.

Definition 3.1.5 (Cocone). A **cocone** C over a diagram $X\colon \mathsf{J} \to \mathsf{C}$ with *nadir* c is a natural transformation $C\colon X \Rightarrow \text{const}_c$ from the constant diagram to X. The components $\{C_j\colon X_j \to c\}_{j\in\mathsf{J}}$ are called the *legs* of the cone.
Less abstractly, a cocone C over $X\colon \mathsf{J} \to \mathsf{C}$ with *nadir* c is a collection of morphisms $C_j\colon X_j \to c$ for every object $j \in \mathsf{J}$ such that for every morphism $f : j \to k$ in J the following diagram in C commutes

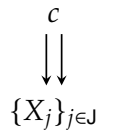


If $C\colon X \Rightarrow \text{const}_c$ and $D\colon X \Rightarrow \text{const}_d$ are two cocones over X a morphism of cocones is a natural transformation $\text{const}_c \Rightarrow \text{const}_d$ making the following diagram commute

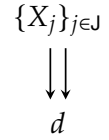


Therefore for any diagram $X\colon \mathsf{J} \to \mathsf{C}$ we have a category of cocones $X \downarrow \Delta$.

Notation. For a diagram $X\colon \mathsf{J} \to \mathsf{C}$ we will denote a cone $C\colon \text{const}_c \Rightarrow X$ by



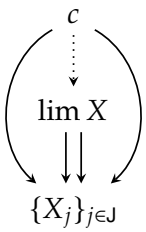
and a cocone $D\colon X \Rightarrow \text{const}_d$ by



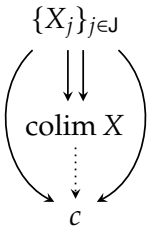
3.2 Limits and colimits

Limits and colimits unify diverse geometric and algebraic constructions by identifying them as the "best possible" approximations of a diagram, formalizing their universal properties.

Definition 3.2.1 (Limit)**.** A **limit** of a diagram $X\colon \mathsf{J} \to \mathsf{C}$ is a terminal object in the category of cones $\Delta \downarrow X$.
Less abstractly, a limit is a cone $\lim X\colon \text{const}_{\lim X} \Rightarrow X$ such that for any other cone $C\colon \text{const}_c \Rightarrow X$ there exists a unique morphism $c \to \lim X$ making the following diagram commute



Definition 3.2.2 (Colimit)**.** A **colimit** of a diagram $X\colon \mathsf{J} \to \mathsf{C}$, denoted $\operatorname{colim} X$, is an initial object in the category of cocones $X \downarrow \Delta$.
Less abstractly, a colimit is a cocone $\operatorname{colim} X\colon X \Rightarrow \text{const}_{\operatorname{colim} X}$ such that for any other cocone $C\colon X \Rightarrow \text{const}_c$ there exists a unique morphism $\operatorname{colim} X \to c$ making the following diagram commute



Remark 3.2.3. Limits and colimits do not need to exist in general for all diagrams and categories.
Categories where all limits and colimits exist are called complete and cocomplete categories respectively. Set is an example of both a complete and cocomplete category, and so are all of the categories we are going to study.
For more information we refer to [**riehlCategoryTheoryContext**].

As a direct consequence of their definition as initial or initial by Proposition 1.3.3 we have that these objects are unique up to categorical isomorphism.

Proposition 3.2.4 (Uniqueness of limits and colimits)**.** If a diagram $X\colon \mathsf{J} \to \mathsf{C}$ has a limit/colimit, it is unique up to unique isomorphism.

Before going into the examples, we fix some terminology around the interaction of

functors with limits and colimits.

Definition 3.2.5 (Functors and limits). Let $F\colon \mathsf{C} \to \mathsf{D}$ be a functor and $X\colon \mathsf{J} \to \mathsf{C}$ a diagram. We say that

- F **preserves** limits if whenever (L, φ) is the limit of X in C, $(F(L), F(\varphi))$ is the limit of $F \circ X$ in D.
- F **reflects** limits if whenever (L, φ) is a cone over X such that $(F(L), F(\varphi))$ is the limit of $F \circ X$ in D then (L, φ) is the limit of X in C.

Replacing cones with cocones and limits with colimits we get dual statements.

3.2.1 Examples of limits and colimits

We now define some special cases of limits and colimits that are generally useful. However, it is important to take into account that this constructions do not exist in general for all categories.

Products and coproducts

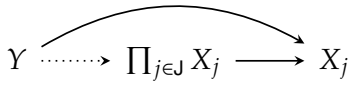
Let J be a small and discrete category.

Since the only morphisms in J are the identity morphisms, a diagram $X\colon \mathsf{J} \to \mathsf{C}$ consists in a collection of objects $\{X_j\}_{j\in\mathsf{J}}$.

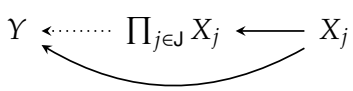
Therefore a cone with apex c over X is an object $c \in \mathsf{C}$ together with morphisms $c \to X_j$ for each $j \in \mathsf{J}$ and a cocone with nadir $c \in \mathsf{C}$ over X is an object c together with morphisms $X_j \to c$ for each $j \in \mathsf{J}$.

Note that any collection of objects $\{X_j\}_{j\in\mathsf{J}}$ in C may be considered as a diagram $X\colon \mathsf{J} \to \mathsf{C}$.

Definition 3.2.6 (Product). Let $\{X_j\}_{j\in\mathsf{J}}$ be a collection of objects in a category C.
We define the **product** of $\{X_j\}_{j\in\mathsf{J}}$, denoted $\prod_{j\in\mathsf{J}} X_j$, as the limit of the diagram $X\colon \mathsf{J} \to \mathsf{C}$.
Less abstractly, a **product** is an object $\prod_{j\in\mathsf{J}} X_j \in \mathsf{C}$ together with maps $\pi_k\colon \prod_{j\in\mathsf{J}} X_j \to X_k$ for every $j \in \mathsf{J}$ such that, for any map $Y \to X_j$ there exists a unique morphism $Y \to \prod_{j\in\mathsf{J}} X_j$ making the following diagram commute



Definition 3.2.7 (Coproduct). Let $\{X_j\}_{j\in\mathsf{J}}$ be a collection of objects in a category C.
We define the **coproduct** of $\{X_j\}_{j\in\mathsf{J}}$, denoted $\coprod_{j\in\mathsf{J}} X_j$, as the colimit of the diagram $X\colon \mathsf{J} \to \mathsf{C}$.
Less abstractly, a **coproduct** is an object $\coprod_{j\in\mathsf{J}} X_j \in \mathsf{C}$ together with maps $\iota_k\colon X_k \to \coprod_{j\in\mathsf{J}} X_j$ for every $j \in \mathsf{J}$ such that, for any map $X_j \to Y$ there exists a unique morphism $\coprod_{j\in\mathsf{J}} X_j \to Y$ making the following diagram commute



Remark 3.2.8. Consider the case where $\mathsf{J} = \varnothing$ is the empty category, i.e: $\mathrm{Ob}(\mathsf{J}) = \varnothing$.
Suppose that products exist in C.
Let's see how are $\varnothing$-shaped products (empty products).

Note that the family of objecs $\{X_j\}_{j\in\varnothing} \subseteq \mathsf{C}$ is empty, and so there are no projection maps $\pi_k : \prod_{j\in\varnothing} X_j \to X_k$ nor morphisms $\{f_j : Y \to X_j\}_{j\in\varnothing}$ for all $Y \in \mathsf{C}$. Therefore the product condition is vacuosly true, i.e: there exists a unique map $Y \to \prod_{j\in\varnothing} X_j$ for all $Y \in \mathsf{C}$. This means that $\prod_{j\in\varnothing} X_j$ is a terminal object in C.
Dually, if C has coproducts then the empty coproduct is initial in C.

Equalisers and coequalisers

Consider the *parallel pair* category

$$\mathsf{J} = \quad \bullet \rightrightarrows \bullet$$

A diagram $X : \mathsf{J} \to \mathsf{C}$ corresponds to a pair of morphisms $f, g : A \rightrightarrows B$ in C.

Therefore a cone $C : \text{const}_c \Rightarrow X$ is a morphism $c : C \to A$ such that $fc = gc$:

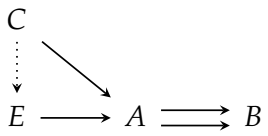
$$C \xrightarrow{c} A \underset{g}{\overset{f}{\rightrightarrows}} B$$

Analogously a cocone $D : X \Rightarrow \text{const}_d$ is a morphism $d : B \to D$ such that $df = dg$:

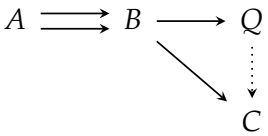
$$A \underset{g}{\overset{f}{\rightrightarrows}} B \xrightarrow{d} D$$

Note that any pair of morphisms $f, g : A \rightrightarrows B$ can be considered as a diagram $X : \mathsf{J} \to \mathsf{C}$.

Definition 3.2.9 (Equaliser). Let $f, g : A \rightrightarrows B$ be two morphisms in C, considered as a diagram $X : \mathsf{J} \to \mathsf{C}$. The **equaliser** of f and g is the limit of this diagram.
Less abstractly, it is an object $E \in \mathsf{C}$ together with a morphism $\text{eq} : E \to A$ such that for any morphism $c : C \to A$ with $fc = gc$ there exists a unique map $C \to E$ the following diagram commute



Definition 3.2.10 (Coequaliser). Let $f, g : A \rightrightarrows B$ be two morphisms in C, considered as a diagram $X : \mathsf{J} \to \mathsf{C}$. The **coequaliser** of f and g is the limit of this diagram.
Less abstractly, it is an object $Q \in \mathsf{C}$ together with a morphism $\text{coeq} : B \to Q$ such that for any morphism $d : B \to C$ with $df = dg$ there exists a unique map $Q \to C$ making the following diagram commute



Proposition 3.2.11. Equaliser arrows are monomorphisms. Dually, coequaliser arrows are epimorphisms.

Proof. Let $\text{eq} : E \to X$ be a equaliser. Consider two morphisms $f, g : C \rightrightarrows E$ in C such that $\text{eq} \circ f = \text{eq} \circ g$. By the uniqueness of the map $\text{eq} : E \to X$ we have $f = g$. □

Example 3.2.12 (Equalisers and coequalisers in Set). In Set the equaliser of two maps $f, g\colon A \rightrightarrows B$ is $E = \{a \in A \mid f(a) = g(a)\}$ together with the inclussion $E \hookrightarrow A$. The coequaliser of f and g is $Q = {}^{B}\!/_{\sim}$ where $\sim$ is the smallest equivalence relation such that for all $a \in A \quad f(a) \sim g(a)$ together with the natural projection

$$\begin{aligned} \mathrm{coeq}\colon B &\twoheadrightarrow Q \\ b &\mapsto [b] \end{aligned}$$

A special case of equalisers and coequalisers are kernels and cokernels.

Definition 3.2.13 (Kernel). Let C be a category with a zero object $0 \in \mathsf{C}$. The **kernel** $\ker f$ of a morphism $f\colon A \to B$ is the equaliser

$$\ker f \rightarrowtail A \underset{0}{\overset{f}{\rightrightarrows}} B$$

Less abstractly, it is an object $\ker f$ together with a morphism $i\colon \ker f \to X$ such that for every $C \in \mathsf{C}$ and every morphism $g\colon C \to X$ such that $f \circ g = 0$ there is a unique morphism $\phi\colon C \to \ker f$ such that $g = p \circ \phi$

Definition 3.2.14 (Cokernel). Let C be a category with a zero object $0 \in \mathsf{C}$. The **cokernel** $\operatorname{coker} f$ of a morphism $f\colon A \to B$ is the coequaliser

$$A \underset{0}{\overset{f}{\rightrightarrows}} B \twoheadrightarrow \operatorname{coker} f$$

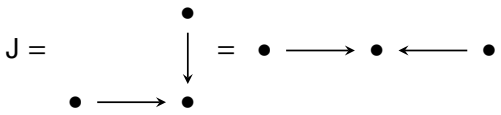
Less abstractly, it is an object $\operatorname{coker} f$ together with a morphism $p\colon Y \to \operatorname{coker} f$ such that for every $C \in \mathsf{C}$ and every morphism $g\colon Y \to C$ such that $g \circ f = 0$ there is a unique morphism $\phi\colon \operatorname{coker} f \to C$ such that $g = \phi \circ p$.
We denote

$${}^{A}\!/_{B} := \operatorname{coker}(A \to B)$$

Remark 3.2.15. By Proposition 3.2.11, kernels are monomorphisms and cokernels are epimorphisms.

Pullbacks and pushouts

Consider the category



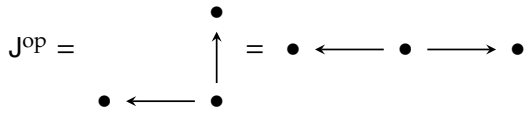
A diagram $X\colon \mathsf{J} \to \mathsf{C}$ consists of a pair of morphisms $A \xrightarrow{f} Z \xleftarrow{g} B$.

Therefore a cone $\mathrm{const}_C \Rightarrow X$ consists of an object $C \in \mathsf{C}$ together with morphisms $C \to X$ and $C \to Y$ making the following diagram commute.

$$\begin{array}{ccc} C & \longrightarrow & A \\ \downarrow & & \downarrow{\scriptstyle f} \\ B & \underset{g}{\longrightarrow} & Z \end{array}$$

Note that any pair of morphisms $A \xrightarrow{f} Z \xleftarrow{g} B$ can be considered as a diagram $X\colon \mathsf{J} \to \mathsf{C}$.

Consider now the opposite category



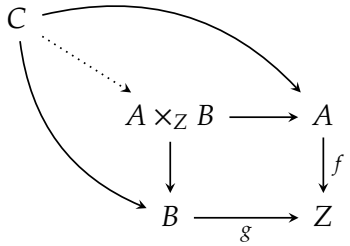
A diagram $Y\colon \mathsf{J}^{\mathrm{op}} \to \mathsf{C}$ consists of a pair of morphisms $A \xleftarrow{f} Z \xrightarrow{g} B$.

Therefore a cocone $X \Rightarrow \mathrm{const}_D$ consists of an object $D \in \mathsf{D}$ together with morphisms $A \to D$ and $B \to D$ making the following diagram commute

$$\begin{array}{ccc} Z & \xrightarrow{f} & A \\ {\scriptstyle g}\downarrow & & \downarrow \\ B & \longrightarrow & D \end{array}$$

Note that any pair of morphisms $A \xleftarrow{f} Z \xrightarrow{g} B$ can be considered as a diagram $X\colon \mathsf{J}^{\mathrm{op}} \to \mathsf{C}$.

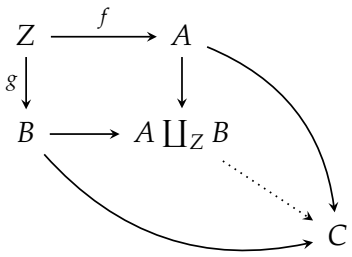
Definition 3.2.16 (Pullback). Let $A \xrightarrow{f} Z \xleftarrow{g} B$ be a pair of morphisms in C, considered as a diagram $X\colon \mathsf{J} \to \mathsf{C}$.
The **pullback** (or **fibre product**) of f and g is the limit of this diagram, denoted by $A \times_Z B$.
Less abstractly, it is an object $A \times_Z B \in \mathsf{C}$ together with a pair of morphisms $\pi_A\colon A \times_Z B \to A, \pi_B : A \times_Z B \to B$ such that for any pair of morphisms $C \to A, C \to B$ making the square commute, there exists a unique morphism $C \to A \times_Z B$ such that



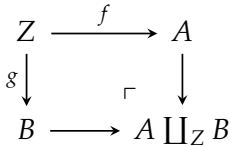
Pullback diagrams are often denoted by

$$\begin{array}{ccc} A \times_Z B & \longrightarrow & A \\ \downarrow & \ulcorner & \downarrow{\scriptstyle f} \\ B & \xrightarrow[g]{} & Z \end{array}$$

Definition 3.2.17 (Pushout). Let $A \xleftarrow{f} Z \xrightarrow{g} B$ be a pair of morphisms in C, considered as a diagram $X\colon \mathsf{J}^{\mathrm{op}} \to \mathsf{C}$.
The **pushout** of f and g is the colimit of this diagram, denoted by $X \coprod_Z Y$.
Less abstractly, it is an object $A \coprod_Z B \in \mathsf{C}$ together with a pair of morphisms $\iota_A\colon A \to A \coprod_Z B, \iota_B : B \to A \coprod_Z B$ such that for any pair of morphisms $A \to C, B \to C$ making the square commute, there exists a unique morphism $A \coprod_Z B \to C$ such that



Pushout diagrams are often denoted by



Example 3.2.18 (Pullbacks and pushouts in Set)**.** In Set the pullback of two morphisms $A \xrightarrow{f} Z \xleftarrow{g} B$ is the set

$$A \times_Z B = \{(a, b) \in A \times B \mid f(a) = g(b)\}$$

together with the projection maps

$$\begin{aligned} \pi_A : A \times_Z B &\to A \\ (a, b) &\mapsto a \end{aligned} \qquad \begin{aligned} \pi_B : A \times_Z B &\to B \\ (a, b) &\mapsto b \end{aligned}$$

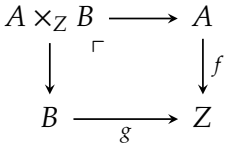
The pushout of two morphisms $A \xleftarrow{f} Z \xrightarrow{g} B$ is the set

$$A \coprod_Z B = {}^{A \amalg B}\!/_{\sim}$$

where $\sim$ is the equivalence relation generated by $f(a) \sim g(a)$ for all $a \in A$ together the inclussion maps

$$\begin{aligned} \iota_A : A &\to A \coprod_Z B \\ a &\mapsto [a, 0] \end{aligned} \qquad \begin{aligned} \iota_B : B &\to A \coprod_Z B \\ b &\mapsto [b, 1] \end{aligned}$$

Remark 3.2.19. Notice how pullbacks in Set can be seen as a equaliser in Set from the product maps $f \circ \pi_A, g \circ \pi_B : A \times B \to Z$. Considering pushouts in Set we get a dual observation.
Indeed, it is not difficult to check that if C has products then the pullback



is equivalent to the equaliser

$$A \times_Z B \longrightarrow A \times B \underset{g \circ \pi_B}{\overset{f \circ \pi_A}{\rightrightarrows}} Z$$

Dually if C has coproducts the pushout

$$\begin{array}{ccc} Z & \xrightarrow{f} & A \\ {\scriptstyle g}\downarrow & \ulcorner & \downarrow \\ B & \longrightarrow & A \coprod_Z B \end{array}$$

is equivalent to the coequaliser

$$Z \underset{\iota_B \circ g}{\overset{\iota_A \circ f}{\rightrightarrows}} A \coprod B \longrightarrow A \coprod_Z B$$

For a proof see [**riehlCategoryTheoryContext**].

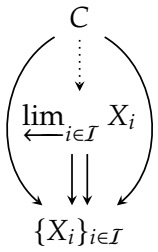
Inverse and direct limits

Definition 3.2.20 (Directed set)**.** A **directed set** is a preordered set $(\mathcal{I}, \leq)$ such that for every $i, j \in \mathcal{I}$, there exists $k \in \mathcal{I}$ with $i \leq k$ and $j \leq k$. We can view this as a category following Example 1.1.11.

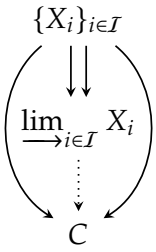
Example 3.2.21. Let $(X, \mathcal{T})$ be a topological space. Then $(\mathcal{T}, \subseteq)$ forms a directed set. This means that the category of open sets $\mathsf{Open}(X)$ can be seen as a directed set.

Definition 3.2.22 (Direct and inverse systems)**.** A **direct system** in C over $\mathcal{I}$ is a diagram $X \colon \mathcal{I} \to \mathsf{C}$. Less abstractly, it consists of a family of objects $\{X_i\}_{i \in \mathcal{I}}$ and morphisms $f_{ij} \colon X_i \to X_j$ for all $i \leq j$, such that $f_{ii} = \mathrm{id}_{X_i}$ and $f_{jk} \circ f_{ij} = f_{ik}$ for all $i \leq j \leq k$.
Dually, an **inverse system** in C over $\mathcal{I}$ is a diagram $X \colon \mathcal{I}^{\mathrm{op}} \to \mathsf{C}$. Less abstractly, it consists of a family of objects $\{X_i\}_{i \in \mathcal{I}}$ and morphisms $p_{ij} \colon X_j \to X_i$ for all $i \leq j$, such that $p_{ii} = \mathrm{id}_{X_i}$ and $p_{ij} \circ p_{jk} = p_{ik}$ for all $i \leq j \leq k$.

Definition 3.2.23 (Inverse limit)**.** The **inverse limit** (or **projective limit**) of an inverse system $X \colon \mathcal{I} \to \mathsf{C}$ is the limit of this diagram, denoted as $\varprojlim_{i \in \mathcal{I}} X_i$.
Less abstractly, it is an object $\varprojlim_{i \in \mathcal{I}} X_i \in \mathsf{C}$ together with morphisms $\pi_i \colon \varprojlim_{i \in \mathcal{I}} X_i \to X_i$ such that $p_{ij} \circ \pi_j = \pi_i$ for all $i \leq j$. For any other object C and morphisms $c_i \colon C \to X_i$ compatible with the system ($p_{ij} \circ c_j = c_i$), there exists a unique morphism $C \to \varprojlim_{i \in \mathcal{I}} X_i$ making the following diagram commute:



Definition 3.2.24 (Direct limit)**.** The **direct limit** (or **inductive limit**) of a direct system $X \colon \mathcal{I} \to \mathsf{C}$ is the colimit of this diagram, denoted as $\varinjlim_{i \in \mathcal{I}} X_i$.
Less abstractly, it is an object $\varinjlim_{i \in \mathcal{I}} X_i \in \mathsf{C}$ together with morphisms $\iota_i \colon X_i \to \varinjlim_{i \in \mathcal{I}} X_i$ such that $\iota_j \circ f_{ij} = \iota_i$ for all $i \leq j$. For any other object C and morphisms $c_i \colon X_i \to C$ compatible with the system ($c_j \circ f_{ij} = c_i$), there exists a unique morphism $\varinjlim_{i \in \mathcal{I}} X_i \to C$ making the following diagram commute:



Example 3.2.25 (Inverse limits in Set). In Set, if $(\{X_i\}_{i\in\mathcal{I}}, \{p_{ij}\}_{i\leq j})$ is an inverse system indexed by a directed set $(\mathcal{I}, \leq)$, its inverse limit is

$$\varprojlim_{i\in\mathcal{I}} X_i = \left\{ (x_i)_{i\in\mathcal{I}} \in \prod_{i\in\mathcal{I}} X_i \mid \forall i \leq j \quad x_i = p_{ij}(x_j) \right\}$$

together with the natural projection maps $\pi_k((x_i)_{i\in\mathcal{I}}) = x_k$.

Example 3.2.26 (Direct limits in Set). In Set, if $(\{X_i\}_{i\in\mathcal{I}}, \{f_{ij}\}_{i\leq j})$ is a direct system indexed by a directed set $(\mathcal{I}, \leq)$, its direct limit is

$$\varinjlim_{i\in\mathcal{I}} X_i = \coprod_{i\in\mathcal{I}} X_i \Big/ \sim$$

where $\sim$ is the equivalence relation defined by $x_i \sim x_j$ if and only if there exists $k \in \mathcal{I}$ with $k \geq i, j$ such that

$$f_{ik}(x_i) = f_{jk}(x_j)$$

together with the natural insertion maps $\iota_k(x_k) = [x_k]$.

3.3 Existence of limits

We finish this chapter by giving a condition for the existence of finite limits and colimits

Theorem 3.3.1. A category has all finite limits if and only if it has a terminal object, binary products and equalisers.
Dually it has all finite colimits if and only if it has a initial object, binary coproducts and coequalisers.

Proof.

$\Longrightarrow$ Direct since all of these are particular cases of limits.

$\Longleftarrow$ Consider a diagram $X : \mathsf{J} \Rightarrow \mathsf{C}$.
Since binary products exist in C we can construct inductively

$$P := \prod_{j\in\mathsf{J}} X_j \in \mathsf{C}$$

For every morphism $u : j \to k$ in J we can consider two maps from P to X_k: the projection $\pi_k : P \to X_k$ and the composition $Xu \circ \pi_j : P \to X_j \to X_k$. Therefore we can consider a pair of parallel morphisms

$$\prod_{j\in\mathsf{J}} X_j \underset{t}{\overset{s}{\rightrightarrows}} \prod_{u:j\to k\in\mathsf{J}} X_k$$

where s is defined in each component as π_k and t is $Xu \circ \pi_j$ in each component. The equaliser of s and t (which exists by hypothesis) is then the limit of X. Indeed, let $\mathrm{eq}\colon L \to \prod_{j\in \mathsf{J}} X_j$ be this equaliser. This comes with a set of morphisms $\{\phi_j\colon L \to X_j\}_{j\in\mathsf{J}}$ such that

$$\phi_j = \pi_j \circ \mathrm{eq}$$

By definition $s \circ \mathrm{eq} = t \circ \mathrm{eq}$, so for every $u\colon j \to k$ in J we have

$$\pi_k \circ \mathrm{eq} = (Xu \circ \pi_j) \circ \mathrm{eq}$$

Therefore

$$\phi_k = Xu \circ \phi_j$$

and so L is a cone over X.
To see that it is a limit cone consider another cone $(W, \{\psi_j\colon W \to X_j\})$ such that for any $u\colon j \to k$ we have $\psi_k = Xu \circ \psi_j$. By definition of the product there exists a unique morphism $h\colon W \to \prod_{j\in\mathsf{J}} W_j$ such that $\pi_j \circ h = \psi_j$.
Note that $s \circ h = t \circ h$. Indeed the components of $s \circ h$ are $\pi_k \circ h = \psi_k$ and the components of $t \circ h$ are $Xu \circ \pi_j \circ h = Xu \circ \psi_j = \psi_k$.
Therefore by the definition of equaliser there exists a unique morphism $\widetilde{h}\colon W \to L$ such that $\mathrm{eq} \circ \widetilde{h} = h$. Finally since for any $j \in \mathsf{J}$

$$\phi_j \circ \widetilde{h} = \pi_j \circ \mathrm{eq} \circ \widetilde{h} = \pi_j \circ h = \psi_j$$

we have that W factors uniquely through L and so L is the limit of X. □

In the particular case of functor categories we have a stronger result: these categories have limits (resp. colimits) if the target category has them.

Proposition 3.3.2. Let C and D be two categories and $\mathsf{Fct}(\mathsf{C},\mathsf{D})$ be its functor category. If D has limits (resp. colimits) of shape J then $\mathsf{Fct}(\mathsf{C},\mathsf{D})$ has limits (resp. colimits) of shape J. Furthermore, they are computed objectwise.

Proof. Consider a J-shaped diagram $X\colon \mathsf{J} \Rightarrow [\mathsf{C},\mathsf{D}]$. We wish to construct a limit object $L \in \mathsf{Fct}(\mathsf{C},\mathsf{D})$ together with a limit cone $\pi_j\colon L \to X_j$ for all $j \in \mathsf{J}$.

For any object $c \in \mathsf{C}$ the *evaluation* functor

$$\begin{array}{ccc} \mathrm{ev}_c\colon [\mathsf{C},\mathsf{D}] & \longrightarrow & \mathsf{D} \\ F & & F(c) \\ \eta \Downarrow & \longmapsto & \downarrow \eta_c \\ G & & G(c) \end{array}$$

allows us to define a diagram $X^c := \mathrm{ev}_c \circ X : J \Rightarrow \mathsf{D}$. By hypothesis there exists

$$\lim_{j\in\mathsf{J}} X^c(j) =: L(c) \qquad \pi^c_j\colon L(c) \to X^c(j)$$

To make L into a functor we need to define its action on morphisms. Consider a morphism $f\colon c \to d$ in C. Since X_j is a functor for all $j \in \mathsf{J}$ we have an induced morphism $X_j f\colon X_j(c) \to X_j(d)$. Let's consider the family of morphism

$$\left\{X_j f \circ \pi^c_j\colon L(c) \to X_j(d)\right\}_{j\in\mathsf{J}}$$

We want to show that this forms a cone over $L(d)$ with apex $L(c)$

Let $u : j \to k$ be a morphism in J. Since $Xu : X_j \Rightarrow X_k$ is a natural transformation we have a commutative diagram

$$\begin{array}{ccc} X_j(c) & \xrightarrow{(Xu)_c} & X_k(x) \\ {\scriptstyle X_j f}\downarrow & & \downarrow{\scriptstyle X_k f} \\ X_j(d) & \xrightarrow[(Xu)_d]{} & X_k(d) \end{array}$$

Since π^c is a limit over X^c we also now that $(Xu)_c \circ \pi^c_j = \pi^c_k$. Therefore we have

$$(Xu)_d \circ (X_j f \circ \pi^c_j) = ((Xu)_d \circ X_j f) \circ \pi^c_j = X_k f \circ (Xu)_c \circ \pi^c_j = X_k f \circ \pi^c_k$$

i.e: it is a valid cone over $L(d)$. By the universal property of $L(d) = \lim_{j \in \mathsf{J}} X^d(j)$ there exists a unique morphism $Lf : L(c) \to L(d)$ such that for all $j \in \mathsf{J}$ the following diagram commutes

$$\begin{array}{ccc} L(c) & \xrightarrow{Lf} & L(d) \\ {\scriptstyle \pi^c_j}\downarrow & & \downarrow{\scriptstyle \pi^d_j} \\ X^c(j) & \xrightarrow[X_j f]{} & X^d(j) \end{array}$$

The functiorality of L follows directly from the uniqueness of this induced map into $L(d)$. Moreover the above commutative square is the naturallity condition for π_j with respect to f. Therefore, for each $j \in \mathsf{J}$ the collection $\{\pi^c_j\}_{c \in \mathsf{C}}$ is a natural transformation $\pi_j : L \Rightarrow X_j$. It is an easy check to see that this collection forms a cone over X in the functor category $\mathsf{Fct}(\mathsf{C}, \mathsf{D})$.

To finish the proof, we need to check the universal property. Consider another functor $M \in \mathsf{Fct}(\mathsf{C}, \mathsf{D})$ equipped with a cone $\tau_j : M \Rightarrow X_j$.

For each $c \in \mathsf{C}$ the evaluation $\tau^c_j := \mathrm{ev}_c \circ M : M(c) \to X^c(j)$ is a cone over X^c. This induces a unique morphism $\phi_c : M(c) \to L(c)$ such that $\pi^c_j \circ \phi_c = \tau^c_j$.

To show that $\phi \equiv \{\phi_c\}_{c \in \mathsf{C}}$ is a natural transformation let $f : c \to d$ be a morphism in C. We want to see that the following diagram commutes

$$\begin{array}{ccc} M(c) & \xrightarrow{\phi_c} & L(c) \\ {\scriptstyle Mf}\downarrow & & \downarrow{\scriptstyle Lf} \\ M(d) & \xrightarrow[\phi_d]{} & L(d) \end{array}$$

Postcompossing with π^d_j we get, on the one hand

$$\pi^d_j \circ (Lf \circ \phi_c) = (\pi^d_j \circ Lf) \circ \phi_c = X_j f \circ \pi^c_j \circ \phi_c = X_j f \circ \tau^c_j$$

And on the other

$$\pi^d_j \circ (\phi_d \circ Mf) = (\pi^d_j \circ \phi_d) \circ Mf = \tau^d_j \circ Mf$$

Since τ_j is a natural transformation we have

$$X_j f \circ \tau_j^c = \tau_j^d \circ M f$$

Therefore since the projections agree for all $j \in \mathsf{J}$ by uniqueness of the maps π_j we get the commutativity of the diagram. Its uniqueness as a morphism of cones follows directly from the objectwise uniqueness of each ϕ_c. □

Chapter 4

More on functors

In this chapter we introduce one of the most fundamental results in category theory: the Yoneda lemma. We also define what an adjoint pair of funtors is and state its relation with limits and colimits.

The main references for this chapter are **[riehlCategoryTheoryContext; maclaneCategoriesWorkingMath aluffiAlgebraChapter02009]**.

4.1 The Yoneda lemma

A common theme in these part of the thesis has been stressing that we want to study objects not by themselves but through its relation with their environment. In particular we can apply these way of thought to categories themselves using functors, and to functors using natural transformations.

Note that every category C has a functor of the form $\mathsf{C} \to \mathsf{Set}$: the (covariant) Hom functor that works by fixing an object $X \in \mathsf{C}$ and sending $Y \in \mathsf{C}$ to its set of morphisms $\mathrm{Hom}(X, Y) \in \mathsf{Set}$.

Therefore we could try to study how a functor $F\colon \mathsf{C} \to \mathsf{Set}$ works on an object $X \in \mathsf{C}$ by studying its relation to $\mathrm{Hom}(X, -)$. This is what is known as the Yoneda lemma.

Instead of giving its proof, we will give an intuitive explanation the philosophy behind it. But first, here is the statement.

Theorem 4.1.1 (Yoneda - **[riehlCategoryTheoryContext]**). Let C be a category, $X \in \mathsf{C}$ an object, and $F\colon \mathsf{C} \to \mathsf{Set}$ a functor. There is a bijection:

$$\Phi : \mathrm{Hom}(\mathrm{Hom}(X, -), F) \xrightarrow{\sim} F(X)$$

given by $\alpha \mapsto \alpha_X(\mathbb{1}_X)$. This bijection is natural in both X and F.

Let's exemplify how this works using the following situation: suppose you are playing a game with a friend that consists on guessing a card he is thinking from a poker deck. Since randomly guessing is boring, you can ask any yes or no question about the card to guide your response. Therefore you could start asking "is the card red?" or "is the number bigger than 5?" until you are only left with one possible option. However, what if you were prohibitted from asking direct questions about the card and instead only could ask about its relation with other cards? In this case

there is also a winning strategy, for example instead of asking if the card is red you may ask "is the card color the same as the king of hearts?". Therefore a card is completely determined by its relation to the other cards.

This is the philosophy behind Yonneda's lemma: objects are completely determined by its environment.

4.2 Adjunctions

Throughout mathematics several constructions appear that are "approximate inverses" of each other. Probably the most recurrent is that of "free" objects constructed from a base space: for example the free group generated by a set is the group consisting of all of the words formed with the elements of the set and their formal inverses. Another example is that of giving a topology to a set by considering the discrete topology. These are both examples of a general phenomenon called adjunction.

Definition 4.2.1 (Adjoint functors). Let $L\colon \mathsf{C} \to \mathsf{D}$ and $R\colon \mathsf{D} \to \mathsf{C}$ be a pair of functors. We say that (L, R) is a pair of **adjoint functors** or that L is left-adjoint to R and R is right-adjoint to L if there exists a natural isomorphism of bifunctors

$$\mathrm{Hom}_{\mathsf{D}}(L(-), -) \cong \mathrm{Hom}_{\mathsf{C}}(-, R(-)) \tag{4.2.1}$$

We represent this diagramatically using

$$\mathsf{C} \underset{R}{\overset{L}{\rightleftarrows}} \mathsf{D}$$

Remark 4.2.2 (Unit and counit of an adjunction). Let (L, R) be a pair of adjoint functors. The natural isomorphism (4.2.1) gives the natural isomorphisms

$$\mathrm{Hom}_{\mathsf{D}}(L(-), L(-)) \cong \mathrm{Hom}_{\mathsf{C}}(-, (G \circ L)(-))$$
$$\mathrm{Hom}_{\mathsf{D}}((L \circ R)(-), -) \cong \mathrm{Hom}_{\mathsf{C}}(R(-), R(-))$$

In particular we get natural transformations

$$\eta\colon \mathbb{1}_{\mathsf{C}} \Rightarrow R \circ L \qquad \varepsilon\colon L \circ R \Rightarrow \mathbb{1}_{\mathsf{D}}$$

that satisfy the *triangular identities*

$$\begin{array}{ccc} L & \overset{L\eta}{\Longrightarrow} & L \circ R \circ L \\ & {\scriptstyle \mathbb{1}_F} \searrow & \Big\Downarrow {\scriptstyle \varepsilon F} \\ & & L \end{array} \qquad \begin{array}{ccc} R & \overset{\eta R}{\Longrightarrow} & R \circ L \circ R \\ & {\scriptstyle \mathbb{1}_R} \searrow & \Big\Downarrow {\scriptstyle F\varepsilon} \\ & & R \end{array}$$

Conversely, if L, R are two pair of functors and ε, η are natural transformations as above then (L, R) form an adjoint pair. For a proof of this see [**riehlCategoryTheoryContext**].

Remark 4.2.3. Left and right adjoints are dual concepts.

Just as we can speak of *the* initial and terminal object in a category we can speak of *the* adjoint of a functor, this time up to natural isomorphism.

Proposition 4.2.4 (Adjoints are unique). The right adjoint (resp. left adjoint) to a functor, provided it exists, is unique up to natural isomorphism

Proof. Let $L\colon \mathsf{C} \to \mathsf{D}$ be a functor.

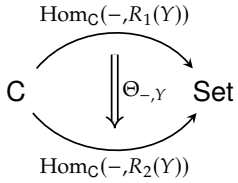
Let $R_1, R_2\colon \mathsf{D} \to \mathsf{C}$ be two functors which are right adjoint to L. This means that for all $X \in \mathsf{C}$ and $Y \in \mathsf{D}$ we have bijections

$$\phi_{X,Y}\colon \operatorname{Hom}_{\mathsf{D}}(L(X), Y) \xrightarrow{\sim} \operatorname{Hom}_{\mathsf{C}}(X, R_1(Y))$$
$$\psi_{X,Y}\colon \operatorname{Hom}_{\mathsf{D}}(L(X), Y) \xrightarrow{\sim} \operatorname{Hom}_{\mathsf{C}}(X, R_2(Y))$$

Using this we can define a bijection

$$\Theta_{X,Y} = \psi_{X,Y} \circ \phi_{X,Y}^{-1}\colon \operatorname{Hom}_{\mathsf{C}}(X, R_1(Y)) \xrightarrow{\sim} \operatorname{Hom}_{\mathsf{C}}(X, R_2(Y))$$

Since ψ and ϕ are natural in X so is Θ. Therefore for a fixed $Y \in \mathsf{D}$ we have a natural isomorphism



By the Yoneda lemma $\Theta_{-,Y}$ induces a unique isomorphism in C

$$\alpha_Y\colon R_1(Y) \xrightarrow{\sim} R_2(Y)$$

which ensemble into a natural isomorphism

$$\alpha\colon R_1 \Rightarrow R_2$$ □

We end the chapter by stating relation between adjoint functors and (co)limits that will prove really useful later on.

Theorem 4.2.5 ([riehlCategoryTheoryContext]). Right adjoints preserve all limits that exist in their codomain.
Dually left adjoints preserve all colimits that exist in their domain.